

Learning goals:

- Understand PCA and how to implement and apply PCA;
- Understand the properties and limitations of PCA;
- Understand how PCA can be used for dimensionality reduction.

Question 1: Refresh your understanding

Refresh your understanding in the following concepts:

- This assignment requires you to understand very carefully the concept of covariance matrices.
- Eigenvectors and eigenvalues: [3Blue1Brown](#)
- Change of basis functions: [3Blue1Brown](#)

Question 2: Properties of PCA

- (a) Write down the objective function of PCA.
- (b) For each variable in the objective function, provide the dimensionality and the conceptual meaning.
- (c) Does PCA have an analytical solution?
- (d) Is PCA a supervised or unsupervised method?

Question 3: PCA step-by-step

List and describe which computation steps are applied to the data when using PCA. You do not have to write the mathematical formulas but describe clearly what each step is doing.

Question 4: Jupyter notebook PCA

Now put the theory into practice by going through the PCA Jupyter notebook.

Question 5: Submitting to Brightspace

For this assignment, please submit the following:

- The Jupyter notebook with your solutions
- A report (pdf extension) with the answers to the theoretical questions. In this report, add a line stating who contributed to what (please, adhere to the code of conduct). Additionally, please add a line on whether, where, and how you used GenAI for this assignment.

Question 6: (Optional) Test your understanding with real exam questions

- (a) PCA
 - 2021-2022 Part 3
 - 2022-2023 Part 5
 - 2024-2025 Part 3
 - 2025-2026 resit Part 5